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# NOWCASTING ECONOMIC ACTIVITY THROUGH TEXT MINING: EVIDENCE FROM ARMENIA

DISCUSSION PAPER DP-2026-01

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# NOWCASTING ECONOMIC ACTIVITY THROUGH TEXT MINING: EVIDENCE FROM ARMENIA

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## ABSTRACT

This paper examines the application of text-mining techniques to economic news data for improving short-term macroeconomic assessment and nowcasting in Armenia. Drawing on international central bank practices, including those of the Bank of England, the Federal Reserve, and the Central Bank of Nigeria, the study highlights the growing role of machine learning and natural language processing (NLP) in extracting policy-relevant signals from unstructured textual data. Using a dataset of over 100,000 economic news articles, the paper applies a combination of word frequency analysis, co-occurrence and clustering methods, and sentiment and uncertainty indicators to identify key economic themes and dynamics. The results show that news-based indicators capture meaningful information about economic activity, particularly in relation to trade, geopolitical developments, and sectoral trends. The findings suggest that text-based measures can complement traditional macroeconomic models and provide timely insights for monetary policy analysis in data-constrained environments.

|                       |                                                                                                                                                                                                       |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
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## I. INTRODUCTION

Text mining is a useful tool for turning unstructured text into structured and measurable information. Since a large share of available data comes in textual form, methods based on machine learning and natural language processing (NLP) help convert qualitative content into quantitative indicators. In economics, this is especially important because it allows researchers and policymakers to extract information from sources such as news, reports, and policy statements in real time.

One of the main challenges in macroeconomic analysis is the delay in official data. Key indicators like GDP are published with a lag and are often revised, which makes it difficult to assess current economic conditions (Blinder et al. (2008)). Text data, in contrast, is available daily and reflects ongoing developments, expectations, and reactions. This makes it a valuable source for nowcasting and short-term forecasting.

There is a growing literature showing that text mining can support central bank analysis, especially in understanding communication and expectations (Bennani and Farvaque (2014)). At the same time, these methods work best when combined with expert judgment. As noted by Bauer (2000), automated tools can identify patterns in large datasets, but expert input is needed to interpret them correctly and ensure economic relevance.

This paper applies standard text-mining techniques to economic news in Armenia. The goal is not to develop new methods, but to use existing tools, such as word frequency analysis, clustering, and sentiment measurement, to extract useful information for macroeconomic monitoring. This approach follows the idea that text mining can help structure large amounts of information and support prediction (Hotho, Nürnberger, and Paaß (2005)).

The results show that economic news in 2023 was strongly focused on external trade, geopolitical developments, and sectoral activity. Terms related to Russia, Turkey, exports, and markets appear frequently and are closely linked to GDP-related discussions. Sentiment analysis indicates that negative tone dominated during the first half of the year, reflecting inflation pressures, global uncertainty, and regional risks. In addition, uncertainty indicators are closely associated with specific sectors such as mining and commodities, suggesting that sectoral shocks played an important role in shaping overall economic perceptions (Baker,

Bloom, and Davis (2016)).

The Armenian case is particularly relevant because the economy is highly exposed to external developments. Many important changes are reflected in news before they appear in official statistics. However, the analysis also faces challenges, especially the lack of Armenian-language NLP tools, which requires translating texts into English and careful preprocessing.

Overall, the study aims to show that economic news can be used to build timely indicators of economic activity. These indicators can complement traditional data and improve short-term economic assessment.

## II. RELATED CENTRAL BANK PRACTICE

Many central banks worldwide, including the Central Bank of Nigeria, the Central Bank of Russia, the Bank of England, the Federal Reserve Bank of San Francisco, the Central Bank of Turkey, and Norges Bank, are currently using machine learning tools in various areas of their activities (Ortiz, Rodrigo, and Turina (2017)). This section describes the models and text-mining techniques applied to the analysis of monetary policy communication and macroeconomic nowcasting (Allahyari et al. (2017)). The resulting insights are used in monetary policy forecasting and programming processes. In the context of big data and natural language processing, machine learning supports the understanding of market expectations, the monitoring of communication effects, and the detection of early warning signals<sup>1</sup>. Common applications include Google News-based inflation tracking, the construction of news-sentiment indices following policy announcements, and the clustering of policy statements using principal component analysis, for example in pre- and post-crisis comparisons (Siklos (2013)).

A large body of research investigates the relationship between central bank communication and macroeconomic outcomes (Acosta (2014)). Shapiro and Wilson (2019),

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<sup>1</sup> Big data platform is a type of IT solution that combines the features and capabilities of several big data application and utilities within a single solution.

for instance, applied text-mining techniques to transcripts and minutes of FOMC meetings as well as speeches by FOMC members over the period 1986–2013. They extracted a measure of monetary policy sentiment and showed that negative sentiment is inversely related to U.S. output growth. Focusing on Japan, Luangaram and Sethapramote (2016) extracted text-based indicators from Bank of Japan monetary policy documents and found that incorporating communication indicators into a structural vector autoregression model increased the response of output to an interest-rate shock. In a related study, Oshima and Matsubayashi (2018) analyzed minutes from the Bank of Japan’s Monetary Policy Meetings between April 2013 and September 2017 using a Latent Dirichlet Allocation model to identify underlying topics, which were found to significantly affect financial market reactions, particularly on publication days (Lahlou (1996)).

Using a computational linguistics approach, Hubert and Fabien (2017) derived monetary policy sentiment scores from policy documents published by the ECB and the Federal Reserve and examined their relationship with the term structure of private short-term interest-rate expectations. Their results indicate that positive shocks to monetary policy sentiment increase private interest-rate expectations in both the United States and the euro area (Tumala and Omotosho (2019), Deerwester et al. (1990)).

### *Bank of England*

The Bank of England developed a new econometric framework to nowcast macroeconomic data subject to revisions and applied it to predict UK GDP growth in real time (Anesti, Galvão, and Miranda-Agrippino, (2020), Amadxarif et al. (2019)). In addition, the Bank has published numerous working papers on the purposes, applications, and techniques of text mining for central banks (Bank of England (2014)). Bholat (2015) demonstrate the additional value central banks can gain from applying various text-mining techniques. They illustrate how these methods can inform policymaking and address key research questions relevant to central banks. The promise of text mining is not merely theoretical but empirically demonstrated. For example, Warsh (2014) cited the text-mining literature (Schonhardt-Bailey (2013), Hansen, McMahon, and Prat (2014)) as an important influence

on his policy recommendation that the Bank disclose more information about its deliberations. Although Warsh acknowledged that studies analyzing millions of spoken words are “daunting and imperfect,” he concluded that text mining has meaningfully advanced understanding of central banks (Warsh (2014)).

More broadly, these contributions show that text mining can serve as a valuable addition to central banks’ analytical toolkit and support the achievement of policy objectives (Greenacre (1984), Greenacre and Hastie (1987)). In this context, researchers assembled a novel dataset of monthly and quarterly indicators covering more than ten years of real-time data vintages. Macroeconomic forecasters often struggle with nowcasting key indicators like GDP, consumption, and investment. Because macroeconomic data are subject to revisions, it is often unclear which release should serve as the forecast target. Early releases allow timely assessment but are incomplete, whereas later estimates are more reliable but published with delay. The choice of forecast target ultimately reflects the forecaster’s implicit loss function. Policy institutions may prefer more mature releases, as these provide a more accurate picture of economic conditions on which policy decisions are based.

Market participants and professional forecasters partially address this trade-off by considering a broad set of predictors, including successive monthly GDP releases within a given quarter (see, for example, Clements and Galvão (2017)). The proposed framework offers a simple characterization of the stochastic process governing revisions as a function of observable variables and enables a detailed evaluation of how incoming data inform forecasts of quarterly GDP growth, the evolution of forecast uncertainty, and expected revisions to early GDP estimates (Eckley (2015)). The findings indicate that the model’s predictions contain information beyond that embedded in earlier statistical office estimates, that predictive intervals are well calibrated, and that real-time UK GDP nowcasts are comparable to those of professional forecasters. Data on production and labor markets, despite substantial publication delays, account for most of the predictability of revisions.

### *Federal Reserve Bank of San Francisco*

The Federal Reserve Bank of San Francisco has published a series of working papers on measuring news sentiment. Shapiro, Sudhof, and Wilson (2020) develop a new time-series measure of economic sentiment based on text analysis of economic and financial newspaper articles from January 1980 to April 2015. Their measure relies on a lexical sentiment model that combines existing lexicons with a newly constructed lexicon designed specifically to capture sentiment in economic news. Using a subset of articles labeled by human evaluators, they demonstrate that the combined model outperforms alternative lexical approaches in predicting human sentiment ratings. The study also provides insights into currently available methodologies for conducting sentiment analysis using textual data (Kearney and Liu (2014)).

Sentiment analysis typically frames the classification of text, whether a word, phrase, or sentence, as a ternary problem (positive, negative, neutral) or as a rating task. Sentiment reflects the speaker's tone or evaluation of a topic, independent of the topic's inherent characteristics. The literature emphasizes two central challenges in sentiment measurement: domain specificity and complexity. Domain specificity refers to the subject matter of the corpus, as words may carry different meanings across contexts. As highlighted by Loughran and McDonald (2011), many words differ in sentiment orientation between general usage and financial contexts; for example, "liability" is neutral in finance but often negative in common language. Effective sentiment tools must therefore be tailored to the relevant domain. Complexity concerns the compositional and contextual nature of language, including features such as negation and multi-word expressions, where sentiment depends on structure rather than isolated word frequencies.

Building on the expanding literature that quantifies sentiment in central bank communications, recent research extends this approach by constructing time-varying sentiment measures for both the European Central Bank and the Federal Reserve and comparing sentiment across communication channels. Using natural language processing techniques and FinBERT, a machine-learning model trained specifically on financial texts (Hilscher, Nabors, and Raviv, (2024)), the authors develop a state-of-the-art sentiment

indicator for published central bank communications. FinBERT, a pre-trained open-source NLP model adapted to financial data, outperforms most alternative methods in financial sentiment analysis (O'Connor et al. (2010)). The analysis focuses primarily on FOMC minutes and ECB press conferences. The results indicate that Federal Reserve and ECB sentiment move closely together. Local projection analysis further shows that ECB sentiment predicts policy interest rates and deviations from the Taylor rule several quarters ahead, while Federal Reserve sentiment also leads ECB rates. Conversely, stock market returns and deflation expectations tend to precede shifts in central bank sentiment (Aase (2011)).

### *The Central Bank of Nigeria*

The Central Bank of Nigeria applied text-mining techniques to analyze its communication strategy over the period 2004–2019 (Tumala and Omotosho (2019)). Because the policy communiqué issued after each Monetary Policy Committee meeting constitutes a key instrument of central bank communication, the authors constructed a corpus of 87 communiqués comprising 123,353 words. After processing the textual data for analysis, they examined the readability, sentiment, and thematic structure of the documents. Although the volume of CBN communication increased substantially over time, suggesting greater monetary policy transparency, the Coleman–Liau readability index indicates that sentence and word structures became more complex, thereby reducing readability. The estimated average monetary policy net sentiment score was –10.5 percent, reflecting the policy uncertainty faced by the MPC during the sample period. The analysis also shows that the dominant topics in the communiqués were shaped by the Bank's policy objectives and the nature of economic shocks prevailing in each period (Amusa, Yahya, and Balogun (2016)).

The study represents a pioneering effort to analyze central bank communication in Nigeria using automated text-mining methods. It highlights scope for future research to examine the empirical relationship between text-based indicators of central bank communication and key macroeconomic variables such as exchange rates, stock market returns, and GDP

growth, as well as the potential usefulness of sentiment-based indicators for improving monetary policy prediction in Nigeria. Results from the estimated topic model further indicate that themes related to monetary policy and macroeconomic stability dominated communication during 2004–2007. Following the oil price increases of 2009–2011 and the relative macroeconomic stability observed during that period, the prominence of topics related to the implementation of policies addressing domestic constraints increased from September 2009, peaking in May 2011.

Overall, the evolution of monetary policy communication topics appears to have been largely influenced by the CBN’s stated policy objectives and by domestic and global shocks affecting the economy in each period. The persistently negative average sentiment score underscores the heightened uncertainty faced by the economy and the MPC’s cautious assessment of macroeconomic conditions and prospects over the sample period.

### *Bank of Russia*

Economic research on text analysis typically focuses on short-term forecasting (nowcasting) of GDP and industrial output growth, inflation, unemployment, sales, and other macroeconomic indicators in Russia (Yakovleva (2018)). The objective of text mining in this context is to construct a high-frequency indicator based on daily news data to assess the dynamics of economic activity. Big Data applications in economics remain relatively new in Russia, with limited existing research, highlighting the novelty of this approach. In particular, Yakovleva (2018) outlines a methodology for calculating a high-frequency indicator of economic activity using news articles collected from online sources.

The need for such an indicator arises from the absence of comparable measures and the delayed publication of official statistics. For example, GDP, the primary measure of economic growth, is released quarterly, typically 1 to 1.5 months after the end of the reporting quarter. This delay restricts the ability to monitor economic developments in real time and to make timely policy decisions. These limitations underscore the relevance of incorporating big data into economic research, a field that is likely to gain increasing importance given the growing demand for timely and data-intensive analytical tools.

### III. DATA AND METHODOLOGY

#### 1. Data Description and Preprocessing

The database used in this study consists of more than 100,000 economic news articles collected from approximately 40–50 websites and published on a daily basis between 2022 and 2023. Although a longer historical archive of economic news covering nearly ten years is available, the empirical analysis focuses primarily on the most recent period, reflecting increased geopolitical volatility and economic uncertainty. This approach ensures that the constructed indicators capture current dynamics relevant for short-term macroeconomic assessment. From this broader dataset, a subset of approximately 1,700 economic news articles published in 2023 is used for the empirical analysis.

Each observation in the dataset includes several structured metadata fields. These include the publication date, language, publisher attribution, and, where available, thematic tags, in addition to the full article text. These metadata fields allow temporal aggregation, source comparison, and thematic filtering within the analytical framework.

The corpus incorporates different types of textual information. The core of the dataset consists of standard news texts, which provide narrative accounts of economic developments, policy announcements, market conditions, and expert commentary. In addition, semi-structured elements such as named entities, institutional references, dates, and locations are embedded within the articles. Although these elements appear within free text, they follow identifiable patterns and can be systematically extracted using entity recognition techniques. Their identification allows for the analysis of key economic actors, sectors, and policy institutions, thereby enriching thematic and contextual analysis.

Where available, user-generated content, including comments and social media references, provides additional qualitative signals reflecting spontaneous public reactions and perceptions. Although such content may be more heterogeneous, it can offer complementary insights into shifts in sentiment, particularly during periods of heightened uncertainty.

Due to the absence of robust Armenian-language NLP tools, all texts were translated into English prior to analysis. Special attention was given to preserving economic meaning

during translation. Armenian and English economic terminology differ in structure and conceptual framing, and direct translation may distort interpretation. Therefore, additional preprocessing and contextual adjustments were applied to ensure consistency and analytical reliability.

## 2. Word Frequency, Proximity, and Co-occurrence Analysis

Word frequency analysis was conducted to identify the most prominent terms in the corpus. Term frequencies were computed and visualized using ranked distributions and word clouds. To improve interpretability, exclusion lists were applied to remove mechanically frequent but economically neutral terms. The resulting frequency distributions provide an overview of dominant economic themes and serve as an initial step in identifying relevant topics.

To examine relationships between economic concepts, co-occurrence analysis was applied. A distance matrix based on joint term appearances was constructed, allowing the measurement of similarity between keywords. Proximity plots were used to visualize these relationships, particularly focusing on the association between selected target variables (e.g., GDP-related terms) and other keywords (Lewis et al. (2004)). This approach helps identify which concepts are most closely linked within the economic narrative (Masawi, Bhattacharya, and Boulter (2014)).

## 3. Thematic Structure and Clustering

Unsupervised learning techniques were employed to uncover latent thematic structures. Hierarchical clustering was used to group keywords according to similarity in their co-occurrence patterns, producing clusters that reflect distinct economic themes. Link analysis complemented this approach by identifying strong pairwise associations between terms, highlighting dominant connections within the data. In addition, correspondence analysis and multidimensional scaling were used to generate spatial representations of relationships between keywords and time periods. Together, these methods provide a

structured view of thematic relationships in the corpus.

#### 4. Sentiment and Uncertainty Analysis

Sentiment analysis was conducted to quantify the tone of the textual data. The approach combines dictionary-based and rule-based methods to classify content into positive and negative categories. Context-specific exclusion lists were applied to improve accuracy (Tetlock (2007)).

In addition to sentiment, uncertainty-related terms were identified and tracked over time. Statistical measures, including chi-square-based indicators and correlation metrics, were used to evaluate the strength of associations between uncertainty-related keywords and the broader corpus. These indicators provide insights into the evolution of economic sentiment and perceived risk.<sup>2</sup>

### IV. RESULTS

#### 1. Word Frequency Analysis and Thematic Signals

The first empirical output of the analysis is a word frequency distribution derived from the news corpus, summarized visually through word clouds and ranked frequency tables. These visualizations highlight the most frequently occurring terms during the sample period and provide an initial overview of dominant economic themes reflected in daily news coverage. To ensure meaningful interpretation, context-specific exclusion lists were applied prior to visualization. Certain high-frequency terms, such as “percent,” “million,” “billion,” “reported,” “according,” and institutional titles, were removed because they appear mechanically in news reporting and do not convey substantive economic information. In addition, particular attention was given to translation-related distortions arising from differences between Armenian and English economic terminology. Words whose

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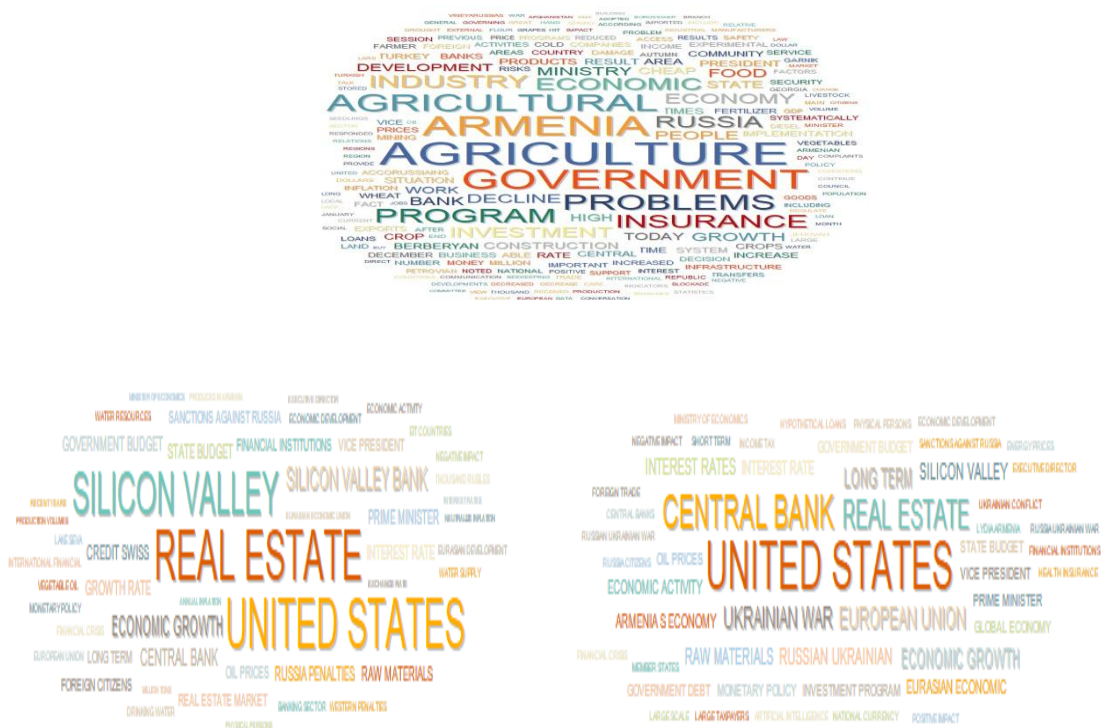
<sup>2</sup> All text mining techniques in this study are implemented using WordStat, which is a text analysis software used to support quantitative content analysis and text mining. It is designed to analyze textual materials such as news articles, interviews, public speeches, survey responses, and electronic communications. For details, please see “WORDSTAT 9 Text Analytics Software User's Guide” Provalis Research, 2021.

translated meaning could bias interpretation were excluded to preserve conceptual consistency.

After filtering, the resulting word clouds (Figure 1) reveal the most prominent economic concepts during the analyzed period. Dominant terms relate to sectors such as real estate, agriculture, government activity, insurance, and external developments involving major economies. These high-frequency themes reflect areas of heightened economic attention and may serve as early signals of sectoral pressures or macroeconomic shifts.

In the context of nowcasting, such frequency-based indicators provide a high-frequency measure of narrative intensity surrounding specific economic topics. While word clouds are primarily exploratory tools, they offer a transparent and intuitive representation of prevailing economic discourse and help identify candidate variables for further modeling and sentiment analysis.

### Figure 1: Word Clouds



**Source:** Daily published economic news, collected and edited by the author.

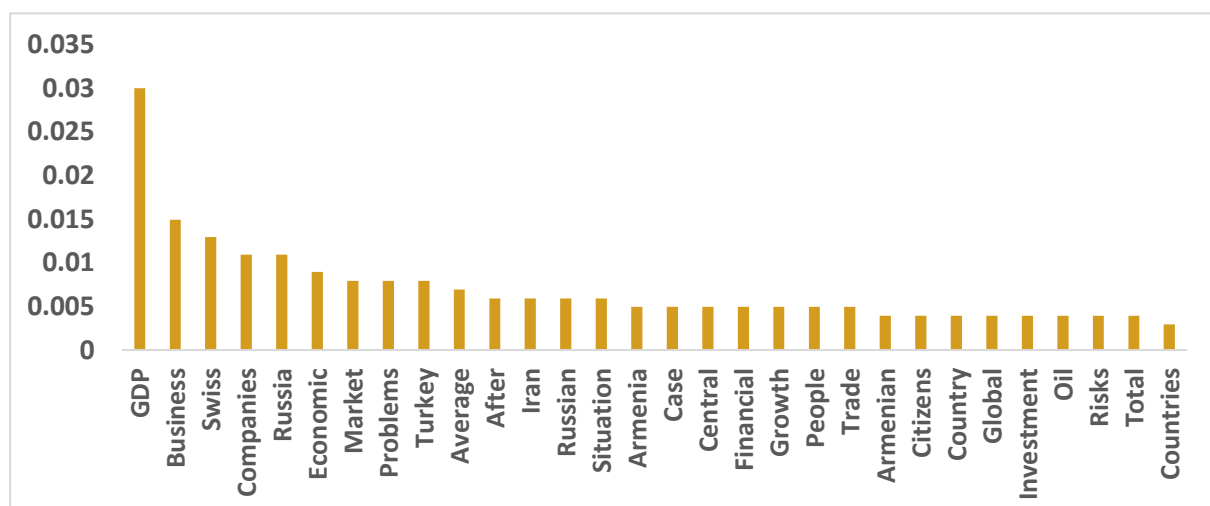
**Notes.** The figure shows the most prominent economic concepts during January-June 2023.

## 2. Proximity Analysis and Thematic Structure

Proximity plots (Figure 2) are used to examine the association between selected target variables and other frequently occurring terms in the corpus. These plots are constructed from the underlying distance matrix and provide a graphical representation of the relative closeness between keywords based on their co-occurrence patterns. Unlike dimensionality reduction techniques, proximity plots serve as exploratory tools that help interpret relationships embedded in high-dimensional textual data.

This analysis uses GDP-related terms as the baseline to pinpoint concepts closely linked to economic activity. The findings show that words like “business,” “companies,” “market,” “economic,” “Russia,” “Turkey,” and external partners often appear near GDP-related terms. This suggests that news narratives linking GDP developments are strongly connected to business activity, external trade relations, and regional geopolitical dynamics. The closeness of these terms implies that fluctuations in discussions surrounding trade partners and corporate activity are central to the media narrative about economic growth. In the Armenian context, where the economy is highly open and externally exposed, such associations highlight the importance of external shocks and regional developments in shaping expectations about output dynamics.

**Figure 2: Proximity plots**



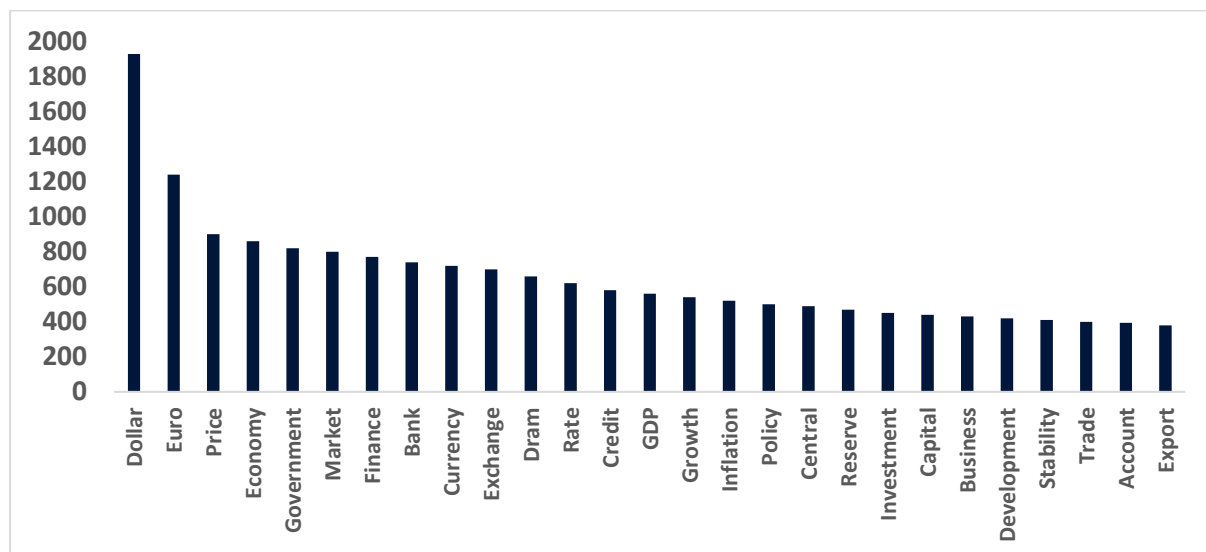
**Source:** Daily published economic news, collected and edited by the author.

**Notes.** The figures show the relationship between the target variables selected according to the economic trends of 2023 and other commonly used terms.

From a nowcasting perspective, proximity analysis helps identify which thematic clusters are most relevant to GDP-related discourse and therefore provides guidance on selecting candidate variables for econometric modeling.

Figure 3 shows keyword frequency distribution in the news corpus. These frequency distributions provide additional insight into the relative intensity of specific economic topics during the sample period. Highly frequent terms reflect areas of concentrated media attention and may indicate sectors or issues that are particularly salient in current economic discourse. Monitoring changes in keyword frequency over time allows the identification of emerging themes and shifts in narrative focus, which are relevant for short-term macroeconomic assessment.

**Figure 3: Distribution of keywords (frequency)**



**Source:** Daily published economic news, collected and edited by the author.

**Notes.** The diagram shows distribution of keyword frequencies across the news corpus, reflecting the higher frequency of use of these topics.

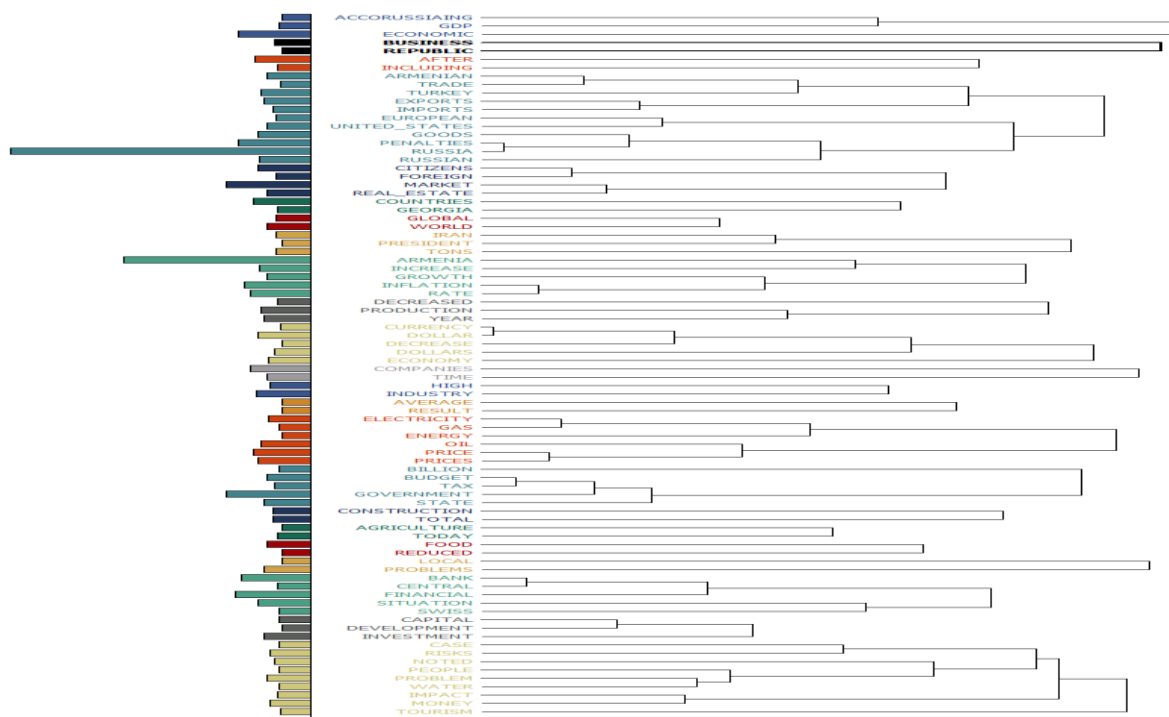
Hierarchical clustering (Figure 4) is used to identify thematic groupings based on co-occurrence patterns among keywords.<sup>3</sup> The results show a clear clustering of terms related to Armenia’s external trade, including “Armenian,” “trade,” “Turkey,” “exports,” and “imports.” Another cluster links terms such as “European,” “US,” “goods,” “penalties,”

<sup>3</sup> Hierarchical Clustering is an unsupervised learning technique that groups data into a hierarchy of clusters based on similarity. This technique groups terms that frequently appear together in news articles, revealing underlying structures in economic narratives.

“Russian,” and “Russia,” suggesting a separate but interconnected narrative centered on sanctions, geopolitical developments, and external economic relations.

The presence of these distinct yet interconnected clusters highlight the strong role of external trade and geopolitical dynamics in shaping economic discussions in Armenia. Given the country’s openness and exposure to regional developments, such thematic structures provide meaningful context for interpreting fluctuations in output and sectoral activity. In terms of nowcasting, clustering findings reveal distinct topic groups that can be combined into composite indicators and used within econometric models.

**Figure 4: Hierarchical clustering**



**Source:** Daily published economic news, collected and edited by the author .

**Notes.** The diagram shows an unsupervised learning technique that groups data into a hierarchy of clusters based on similarity.

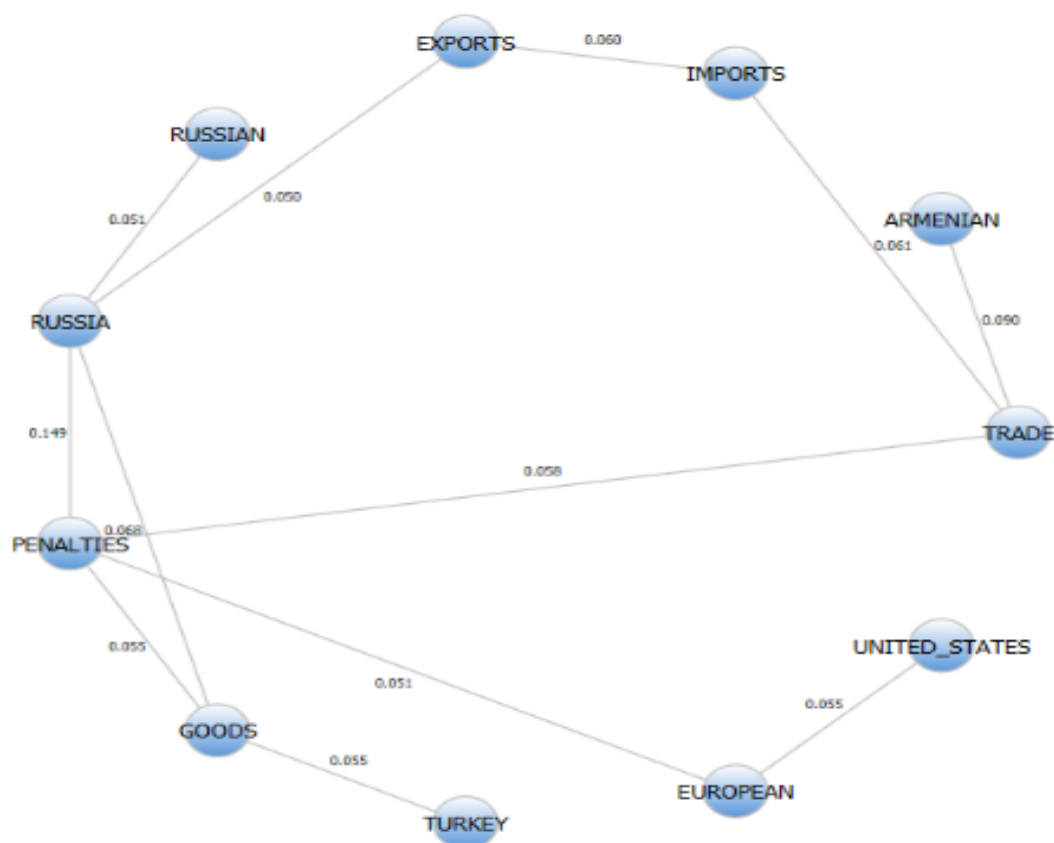
Link analysis (Figure 5) reveals strong co-occurrence between terms related to Russia, sanctions, and goods, indicating that sanctions-related developments constitute a dominant narrative in economic reporting. In contrast, the association between Armenian trade and sanctions appears weaker, suggesting that while external shocks are central in media coverage, Armenia is not always explicitly mentioned as the center of influence of sanctions.

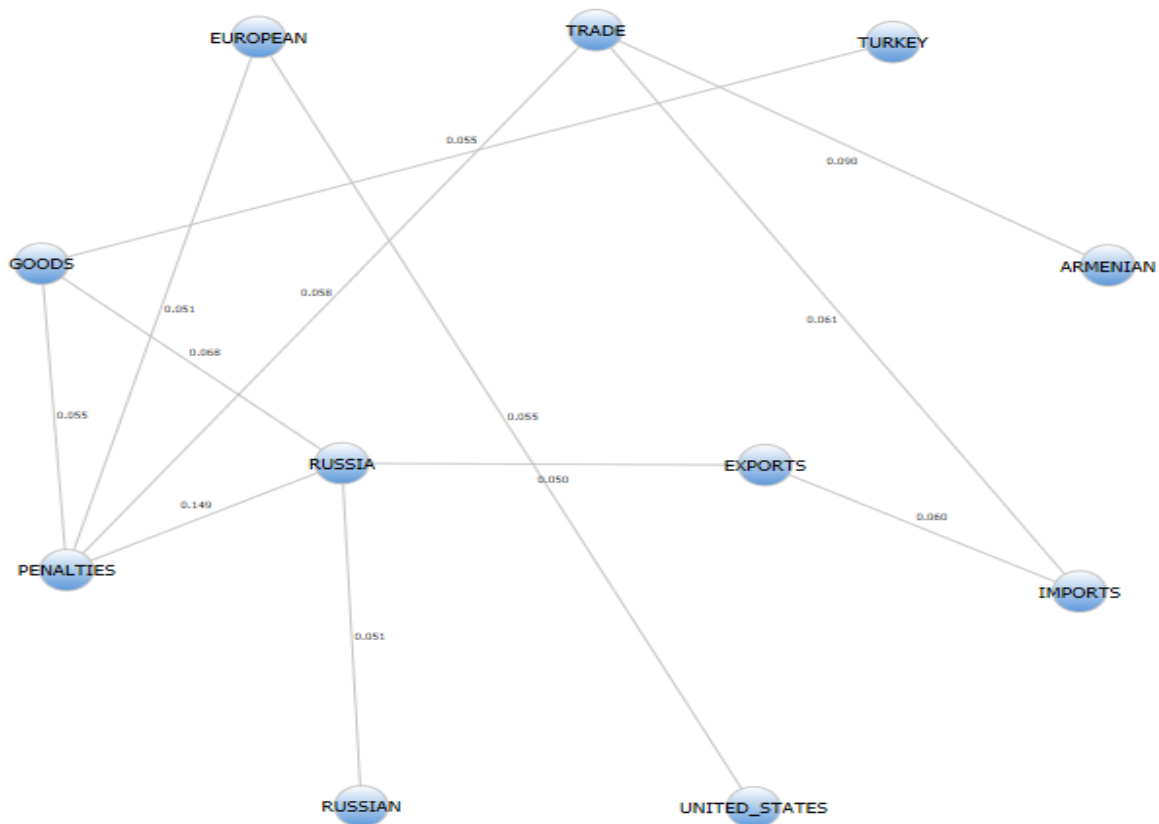
These findings reinforce earlier clustering results and highlight the importance of geopolitical developments in shaping economic expectations. For a small open economy, the prominence of external risk narratives may serve as an early signal of transmission channels affecting domestic output and trade.

Correspondence analysis (Figure 6) provides a spatial representation of relationships among keywords and time periods. The results indicate that certain terms, such as those related to exports, factories, and regional partners, are closely associated with specific months, particularly May 2023. This suggests that sector-specific developments played a prominent role during that period.

This spatial representation confirms the presence of coherent thematic clusters and reinforces the interpretation that trade, sanctions, and sectoral dynamics were central drivers of economic discourse in 2023.

**Figure 5: Link analysis**

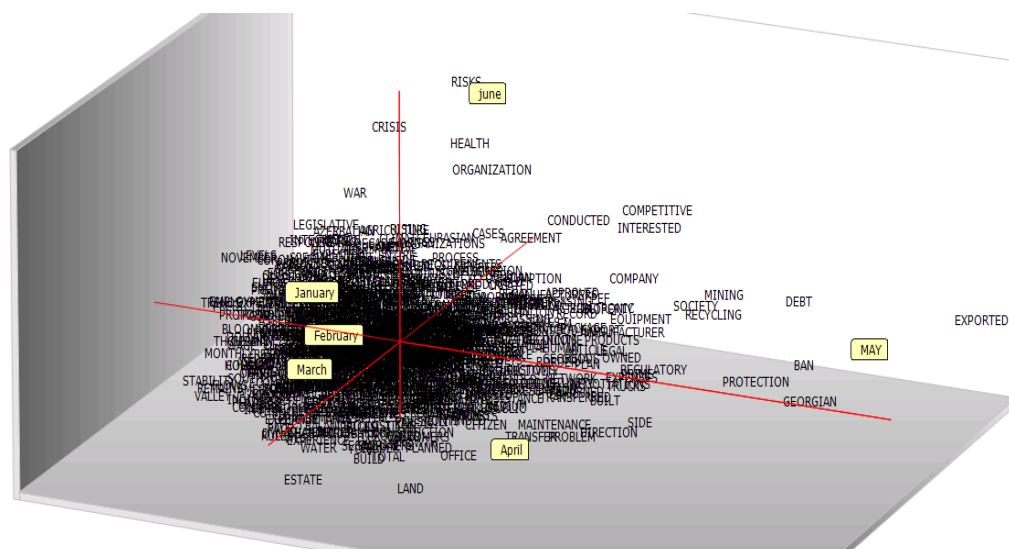




**Source:** Daily published economic news, collected and edited by the author.

**Notes.** The diagram shows which terms and the dominant economic news have a close relationship.

**Figure 6: Correspondence 3D plot**



**Source:** Daily published economic news, collected and edited by the author.

**Notes.** The diagram shows that multidimensional scaling and correspondence relevance analysis provide a spatial representation of relationships among keywords and time periods.

### 3. Sentiment and Uncertainty Indicators

Sentiment analysis results (Figures 7–8) indicate that negative economic tone prevailed between January and June 2023, reflecting global recession concerns, inflationary pressures, and regional geopolitical uncertainty. Dominant themes during this period include inflation, currency movements, and banking stability.

The sentiment indicators display alignment with observed macroeconomic developments, suggesting that news-based measures capture contemporaneous economic conditions and may provide early signals of cyclical shifts. However, the analysis also reveals limited public engagement with official monetary communication, with stronger responses concentrated among financial institutions and professional economists (Apel and Grimaldi (2012)).

Context-specific exclusion lists significantly improved the precision of sentiment measurement by removing mechanically frequent but economically neutral terms.

Figure 9 illustrates the evolution of uncertainty-related themes in 2023. The results suggest that uncertainty was most strongly associated with developments related to Amulsar, commodity prices, and metals during the first half of the year. These sector-specific concerns appear to have contributed to broader macroeconomic risk perceptions.

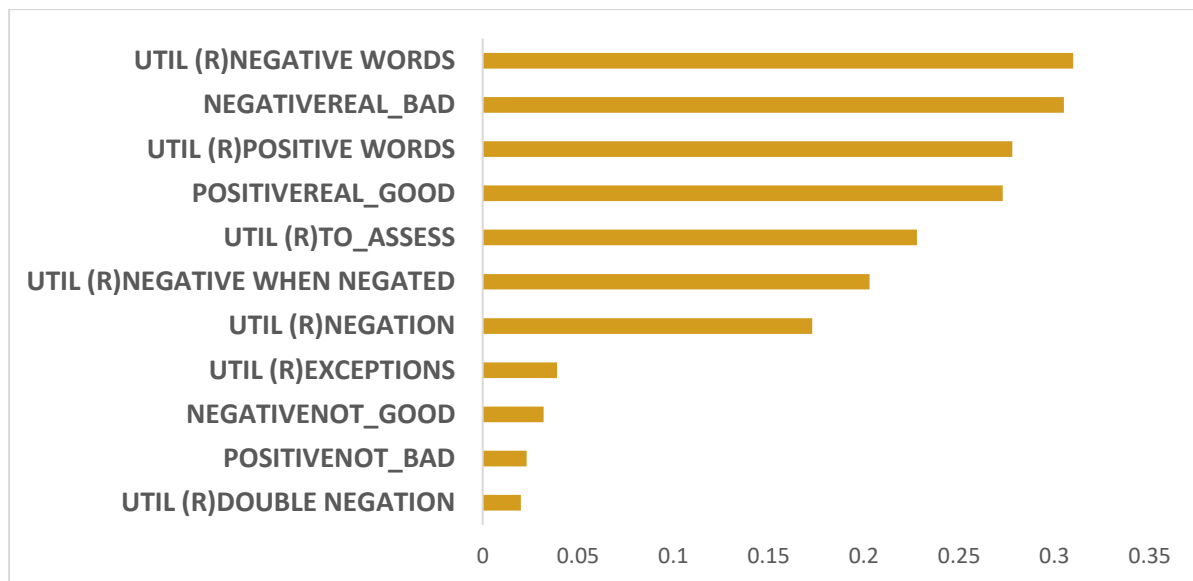
The statistical relationships observed across content categories support the interpretation that external shocks and sectoral volatility were key drivers of economic uncertainty during the sample period.

**Figure 7: Frequency of positive and negative sentiments**



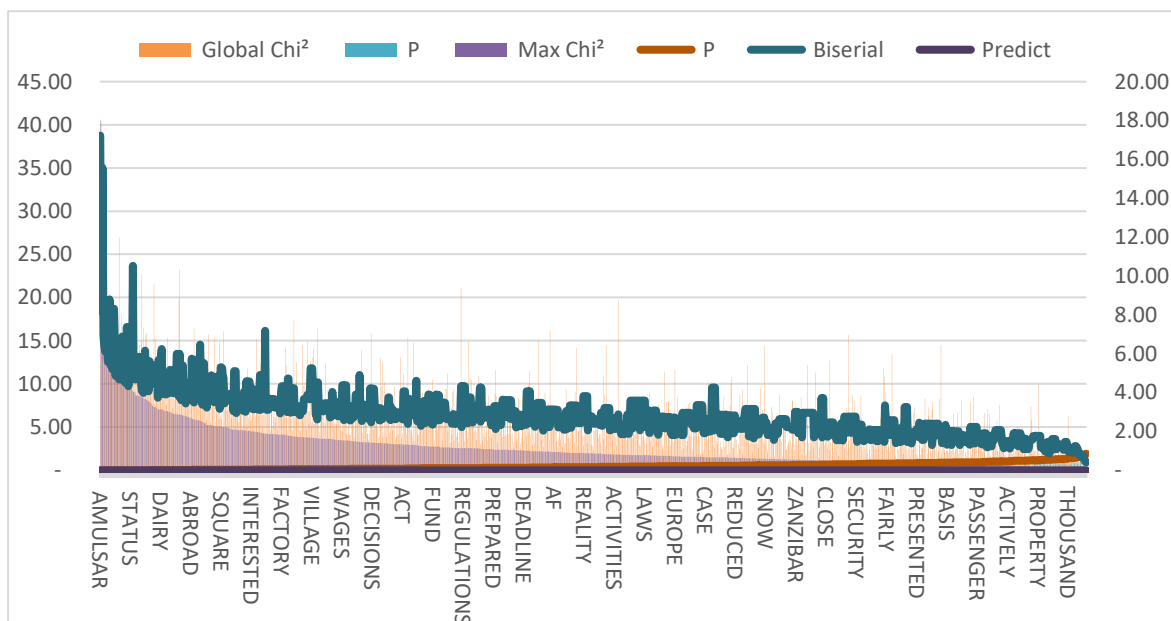
**Source:** Daily published economic news, collected and edited by the author.

**Notes.** The diagram shows that sentiment analysis results indicate that negative economic keywords prevailed between January and June 2023.

**Figure 8: Contribution of sentiment rules**

**Source:** Daily published economic news, collected and edited by the author .

**Notes.** The diagram shows that sentiment analysis results indicate that negative economic tone prevailed between January and June 2023.

**Figure 9: Uncertainties in January-June 2023**

**Source:** Daily published economic news, collected and edited by the author .

**Notes.** The diagram shows the evolution of uncertainty-related themes in January-June 2023. These sector-specific concerns appear to have contributed to broader macroeconomic risk perceptions. The left vertical axis represents the Global Chi-square ( $\chi^2$ ) and Maximum Chi-square ( $\chi^2$ ) statistics, while the right vertical axis corresponds to the association measures, including Biserial correlation, Predict, and P values.

## V.CONCLUSION

This paper demonstrates that text-mining techniques applied to economic news can provide valuable, high-frequency insights into macroeconomic dynamics and complement traditional forecasting approaches. Consistent with the broader literature on central bank communication and text analysis, the findings confirm that textual indicators, such as sentiment and thematic structure, contain meaningful information about economic conditions and expectations (Kent (2014)). In line with Bennani and Farvaque (2014), the results also highlight the practical relevance of text-based methods for central banks, particularly in enhancing policy analysis and communication monitoring.

Empirically, the analysis shows that dominant economic narratives in Armenia during 2023 were closely linked to external trade, geopolitical developments, and sectoral dynamics. Negative sentiment and elevated uncertainty indicators align with global and regional shocks, reinforcing the idea that news-based measures can serve as early signals of economic stress. These findings are consistent with international evidence suggesting that textual sentiment indicators can predict macroeconomic variables and financial market reactions (Shapiro, Sudhof, and Wilson (2020); Hilscher, Nabors, and Raviv (2024)). A key challenge for implementing text mining in macroeconomic monitoring in Armenia is the absence of robust Armenian-language NLP tools, which necessitates translation of texts into English prior to analysis. This increases reliance on expert-driven translation and manual keyword filtering to preserve economic meaning and ensure analytical consistency. Investment in the development of Armenian-language NLP resources would likely improve automation, reduce translation bias, and enhance the scalability of text-based analysis.

Building on this, text-derived indicators could be systematically incorporated into macroeconomic monitoring and forecasting frameworks. The analysis shows that news-based measures capture meaningful shifts in sentiment, uncertainty, and thematic focus that align with observed economic developments, particularly regarding trade, sectoral dynamics, and geopolitical events. Integrating these indicators into nowcasting models would strengthen policymakers' ability to detect emerging risks and turning points, while

continuous updating of dictionaries and exclusion lists would help ensure that the indicators remain relevant and responsive to evolving economic conditions.

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